

Month and year: May 2025

Program: B.Tech. (CS, IT, AI&ML, AI&DS, M&C, R&A)

Subject: Digital System Design (Regular/Reappear)

Maximum Marks: 50

Number of Questions to be attempted: 10

Total no. of pages used: 2

Semester: 2nd

Course code: CSIC100/CSIC12

Time allowed: 3:00 Hrs.

Total No of Questions: 10

Note: 1. There are ten questions in the paper and each is to be attempted.
2. Calculator is allowed.

Q1: (a) Show that the dual of the exclusive-OR is also its complement. [4 X 1]
(b) Perform the following division in binary: $111011 \div 101$.
(c) Convert the following number $(4D.56)_{16}$ to Binary.
(d) Explain the functionality of a Multiplexer.

Q2: (a) Show that a positive logic NAND gate is a negative logic NOR gate and vice versa. [3 X 2]
(b) Perform the following Using BCD arithmetic $(7129)_{10} + (7711)_{10}$.
(c) Implement the full adder using two half adders.

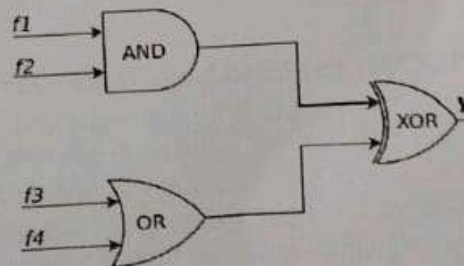
Q3: Add and multiply the following numbers without converting them to decimal. [4]
(i) Binary numbers 1011 and 101.
(ii) Hexadecimal numbers 2E and 34.

Q4: Represent the decimal number 6,248 in (a) BCD, (b) excess-3 code, (c) 2421 code, and (d) a 6311 code. [4]

Q5: Implement the following Boolean function F, using the two-level forms of logic (a) NAND-AND, (b) AND - NOR, (c) OR - NAND, and (d) NOR - OR: [4]
 $F(A, B, C, D) = \sum (0, 4, 8, 9, 10, 11, 12, 14)$

Q6: Simplify the following Boolean functions using four-variable maps. [5]
 $F(A, B, C, D) = \sum (0, 2, 4, 5, 6, 7, 8, 10, 13, 15)$

Q7: Consider 4-variable functions f_1, f_2, f_3, f_4 expressed in sum-of-minterms form as given below. [5]
 $f_1 = \sum (0, 2, 3, 5, 7, 8, 11, 13)$
 $f_2 = \sum (1, 3, 5, 7, 11, 13, 15)$
 $f_3 = \sum (0, 1, 4, 11)$
 $f_4 = \sum (0, 2, 6, 13)$



With respect to the circuit given above, Find out the value of output Y.

Q8: Simplify the function $F(a, b, c, d) = \sum m(2, 3, 4, 5, 13, 15) + dc(8, 9, 10, 11)$. Use 'D' as a MEV variable. [6]

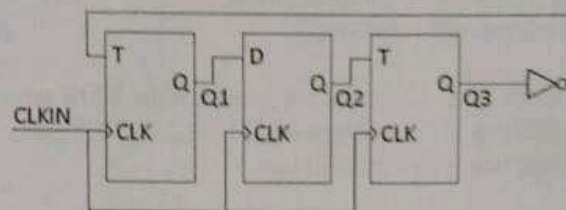
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 Department of Computer Engineering
 End Term Examination (Even Semester, 2024-25)

Q9: Design a sequential circuit with two JK flip-flops A and B and two inputs E and x. If $E = 0$, the circuit remains in the same state regardless of the value of x. When $E = 1$ and $x = 1$, the circuit goes through the state transitions from 00 to 01 to 10 to 11 back to 00, and repeat. When $E = 1$ and $x = 0$, the circuit goes through the state transitions from 00 to 11 to 10 to 01 back to 00, and repeat.

OR

[6]

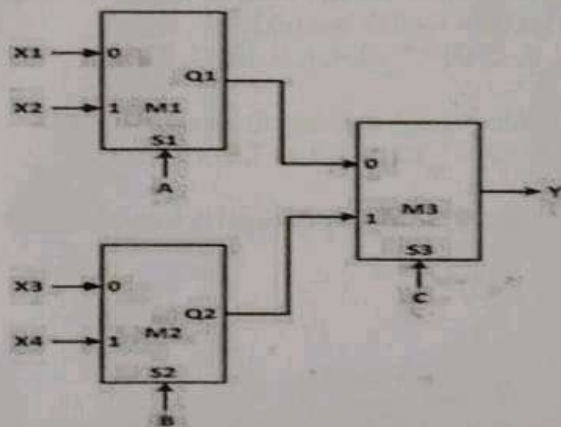
Consider a sequential digital circuit consisting of T flip-flops and D flip-flops as shown in the figure. CLKIN is the clock input to the circuit. At the beginning, Q1, Q2 and Q3 have values 0, 1, 1, respectively.



Find out that which of the given values of (Q1, Q2, Q3) can NEVER be obtained with this digital circuit?

Q10: Consider a digital logic circuit consisting of three 2-to-1 multiplexers M1, M2, and M3 as Shown below. X1 and X2 are inputs of M1. X3 and X4 are inputs of M2. A, B, and C are select lines of M1, M2, and M3, respectively.

[6]



For an instance of inputs $X1=1$, $X2=1$, $X3=0$, and $X4=0$, find out the number of combinations of A, B, C that give the output $Y=1$.